

DSCI 725 Project Analysis

Data Mining for Competitive Advantage

Forecasting Total Deposits at U.S. Commercial Banks

Awuonda Rubby

May 2026

EXECUTIVE SUMMARY

This project developed a univariate time series forecasting model for total deposits held at U.S. commercial banks using monthly data from the Federal Reserve Economic Data (FRED) database (series DPSACBW027SBOG). The objective was to identify the forecasting model with the best predictive accuracy and generate a 24-month forecast to support financial planning and liquidity management.

Initial visual analysis revealed a strong upward trend with limited seasonality. The Augmented Dickey Fuller test confirmed non stationarity in the original series. First differencing achieved statistical stationarity, but ACF inspection suggested remaining persistence. Second differencing produced stronger stationarity and improved autocorrelation behavior and was therefore adopted.

Six forecasting model families were evaluated using an approximately 80/20 chronological training and validation split. These included mean, naïve, seasonal naïve, regression, ETS, and ARIMA models. Forecast accuracy was evaluated using ME, RMSE, MAE, MPE, MAPE, MASE, and ACF1 on both training and validation data.

The ARIMA $d = 2$ specification achieved the best overall validation performance, producing the lowest forecast errors across key accuracy measures while only marginally outperforming ETS and outperforming benchmark models. The final ARIMA(2,2,1)(1,0,1)[12] model was refit on the full dataset and used to generate a 24-month forecast from March 2026 to February 2028. Results indicate continued deposit growth with widening prediction intervals reflecting increasing uncertainty at longer forecast horizons.

INTRODUCTION

Total deposits at U.S. commercial banks represent the primary funding source of the banking system and an important indicator of financial liquidity and economic activity. Growth in deposits increases lending capacity, while declines may indicate tightening financial conditions or reduced economic activity. Accurate forecasting of deposit levels therefore supports lending decisions, liquidity planning, and financial risk management.

This project applies univariate time series forecasting methods to monthly deposit data from January 1973 through early 2026 obtained from the Federal Reserve Economic Data (FRED) database. The analysis follows a structured workflow including data exploration, stationarity testing, model development, model comparison, validation testing, residual diagnostics, and forecasting.

The objective is to identify the forecasting model with the strongest predictive accuracy and produce a 24-month forward forecast.

BACKGROUND RESEARCH

Time series forecasting techniques such as ARIMA and exponential smoothing (ETS) are widely used in economic and financial forecasting because they effectively capture trend, seasonality, and autocorrelation patterns without requiring external predictors. Hyndman and Athanasopoulos (2021) describe these approaches as highly effective across many real-world applications, including macroeconomic indicators and banking aggregates.

Studies applying univariate time series models to aggregate U.S. commercial banking data have shown that ARIMA and ETS approaches produce reliable forecasts when non-stationarity and temporal dependencies are properly addressed through appropriate differencing and model selection. ARIMA models combine autoregressive, differencing, and moving average components to model stationary series, while seasonal ARIMA extends this capability to capture yearly patterns. Exponential smoothing models estimate level and trend using weighted historical observations. Naïve and seasonal naïve models serve as essential baselines since any useful forecasting model must meaningfully outperform simple persistence forecasts.

This methodological foundation, grounded in established forecasting principles, supports the univariate ARIMA and ETS approaches applied in the current project for predicting total deposits at U.S. commercial banks.

PROBLEM PRESENTATION

The objective of this project is to determine which univariate time series forecasting model best predicts total deposits at U.S. commercial banks and to generate a 24-month forward forecast.

The research questions addressed are:

- Which model provides the best predictive accuracy on validation data?
- How are deposits expected to evolve over the next 24 months?
- How does performance compare across forecasting techniques?

Answering these questions supports understanding future banking sector liquidity trends and improves financial planning decisions.

SPECIFICATION AND DESIGN

The modeling workflow followed standard forecasting design practices to ensure valid model selection. Major steps in the modeling workflow included:

- Data preparation
- Stationarity testing

- Model estimation
- Model comparison
- Residual diagnostics
- Forecast generation

Data was split into training and validation sets to evaluate predictive performance on unseen observations and reduce the risk of overfitting. Models were selected based on validation performance rather than training fit to ensure the final model generalizes well to future observations.

DATA ACQUISITION

The data used in this project were obtained from the Federal Reserve Economic Data (FRED) database maintained by the Federal Reserve Bank of St. Louis. The series `DPSACBW027SBOG` represents total deposits at U.S. commercial banks measured monthly in billions of U.S. dollars from January 1973 through early 2026. The dataset was downloaded directly from the FRED database.

The dataset was imported into R as a CSV file for analysis. The time series contained one missing value, which was replaced using linear interpolation to maintain continuity of the series and ensure compatibility with time series modeling techniques.

The data were converted into a monthly time series object with a frequency of 12 to reflect the monthly structure of the observations, as illustrated in Figure 1.

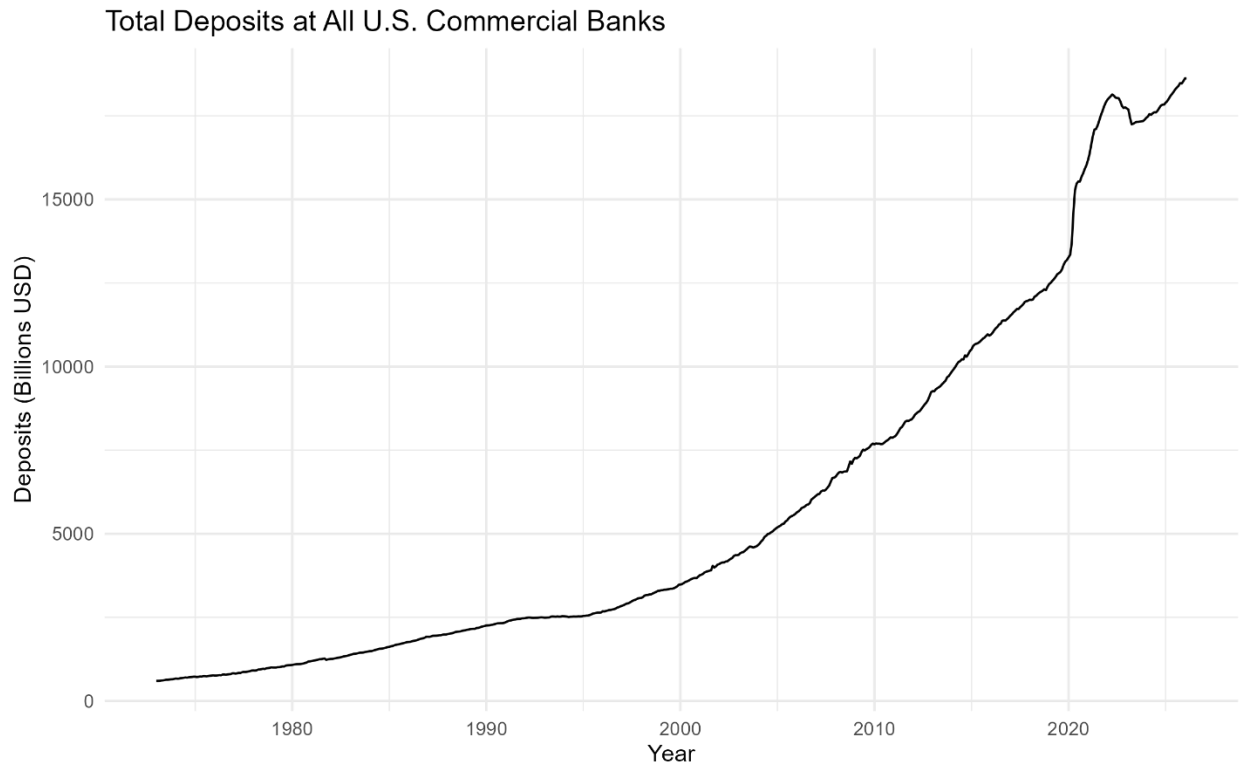


Figure 1. Total Deposits at All U.S. Commercial Banks

DATA EXPLORATION

The time series shows a strong long term upward trend with large increases following the 2008 financial crisis and the 2020 pandemic period. Seasonal variation appears small relative to the dominant trend. These characteristics suggest non stationarity due to trend behavior, making differencing necessary before model estimation.

Augmented Dickey Fuller Test

The Augmented Dickey-Fuller (ADF) test was conducted on the original series, the first-differenced series, and the second-differenced series to formally evaluate stationarity. The purpose was to identify the minimum level of differencing needed for stationarity and to support later model comparison.

Table 1. Augmented Dickey Fuller Test Results

Series	ADF Test Statistic	p-value	Conclusion
Original series	-0.607	0.977	Non-stationary

First-differenced series	-5.327	0.010	Stationary
Second-differenced series	-14.129	0.010	Stationary

The original series produced a p-value of approximately 0.977, indicating non-stationarity. First differencing achieved stationarity with a p-value near 0.01, meaning first differencing was sufficient based on the ADF test. Second differencing also produced a stationary series. Because the ADF test alone does not determine the best forecasting model, both first-differenced and second-differenced ARIMA specifications were later evaluated using validation accuracy and residual diagnostics.

ACF Analysis

ACF of First Differenced Deposits Series

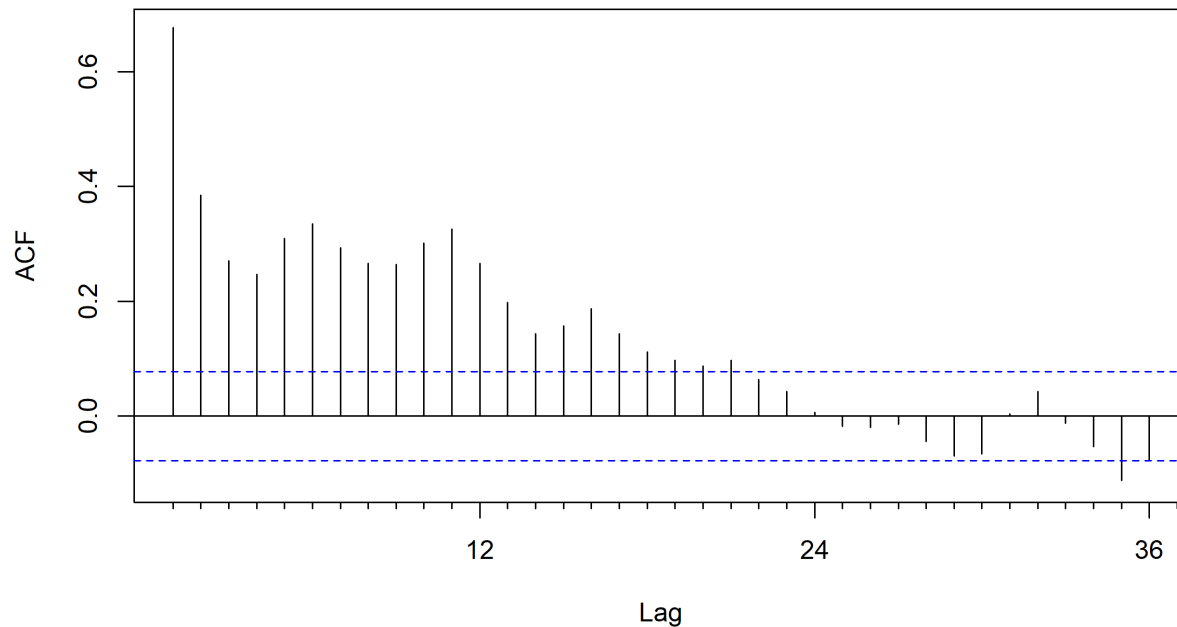


Figure 2. ACF of First Differenced Deposits Series

Figure 2 shows the autocorrelation function of the first differenced series. Although the series achieves stationarity, strong positive autocorrelation persists across several early lags, followed by a gradual decay. This pattern indicates that significant dependence remains in the data even after first differencing. A noticeable seasonal spike appears at lag 12, suggesting annual seasonality. Overall, the slow decay and persistent autocorrelation suggest that first differencing alone may not fully capture the underlying structure of the series.

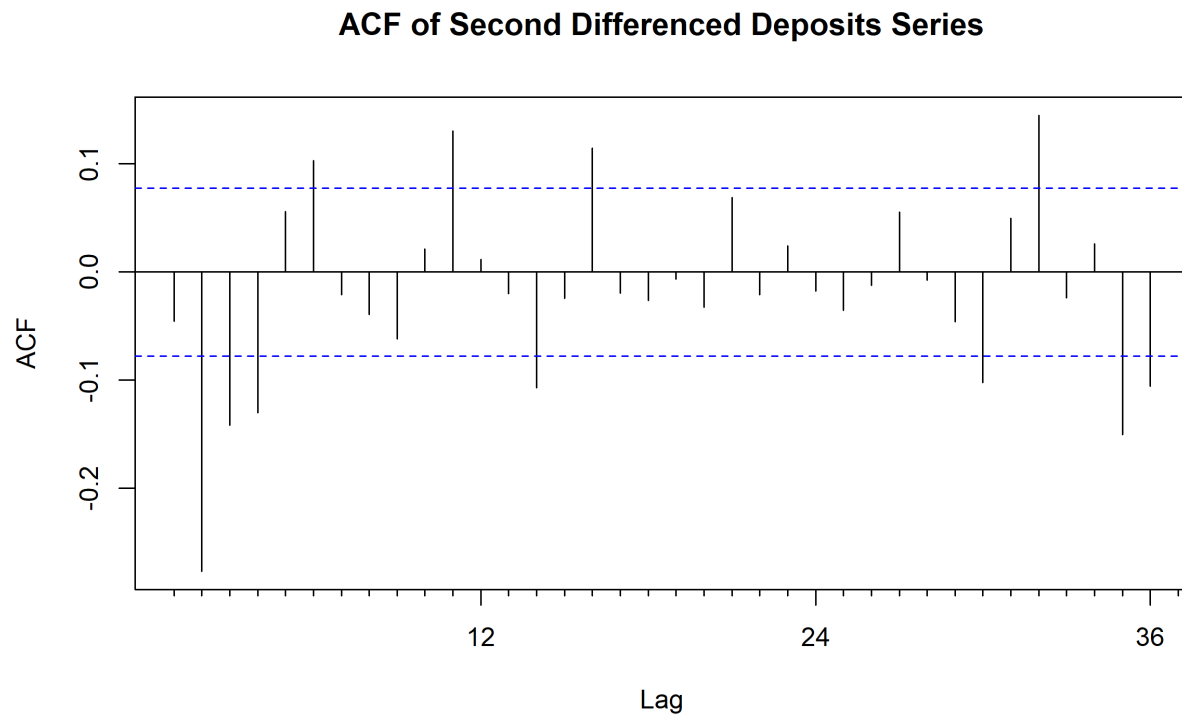


Figure 3. ACF of Second Differenced Deposits Series

Figure 3 shows the autocorrelation function of the second differenced series. The ACF displays a significant negative spike at lag 1 and a smaller spike at lag 2, followed by rapid decay within the confidence bounds. A seasonal spike appears at lag 12, suggesting annual seasonality.

This pattern indicates the presence of a moving average process rather than a purely autoregressive process. The early spikes suggest MA(1) behavior, while the seasonal spike supports inclusion of a seasonal MA(1) component. These observations supported testing ARIMA models that include moving average components.

PACF Analysis

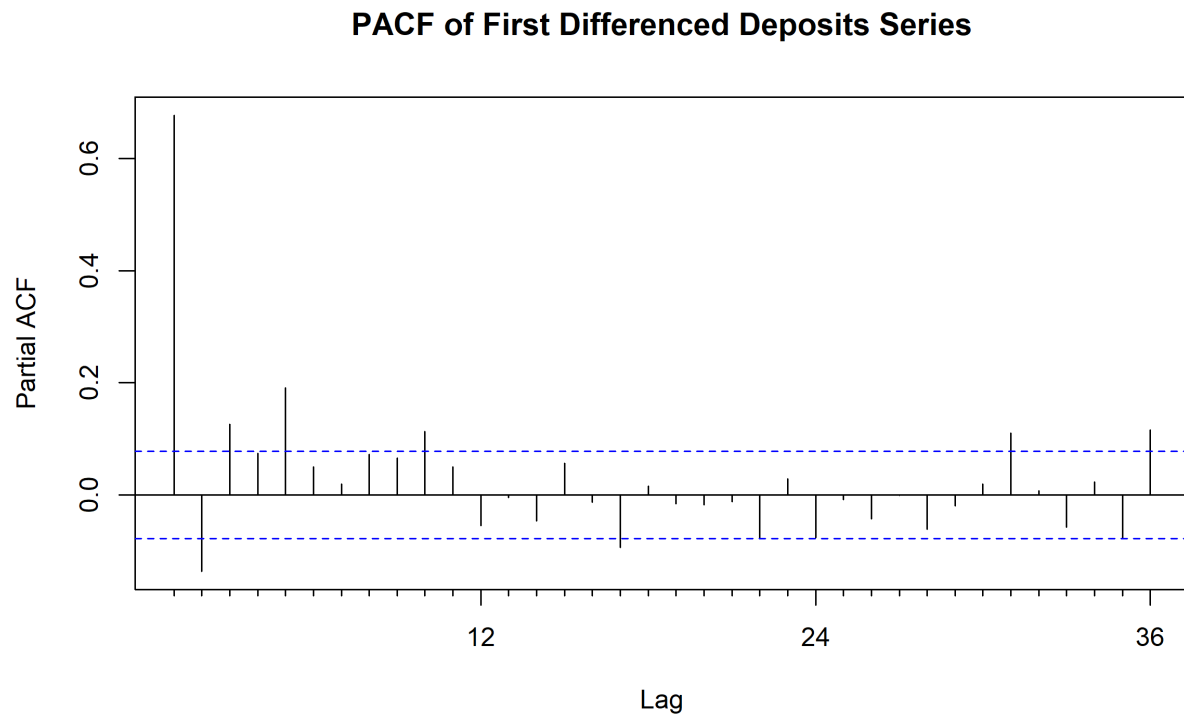


Figure 4. PACF of First Differenced Deposits Series

Figure 4 shows the partial autocorrelation function of the first differenced series. A strong positive spike appears at lag 1, followed by a rapid drop-off in subsequent lags. This pattern suggests the presence of an autoregressive component, specifically AR(1) behavior. However, smaller spikes remain at later lags, indicating that the series is not fully white noise after first differencing. Together with the ACF results, this suggests that first differencing does not fully eliminate dependence on the data.

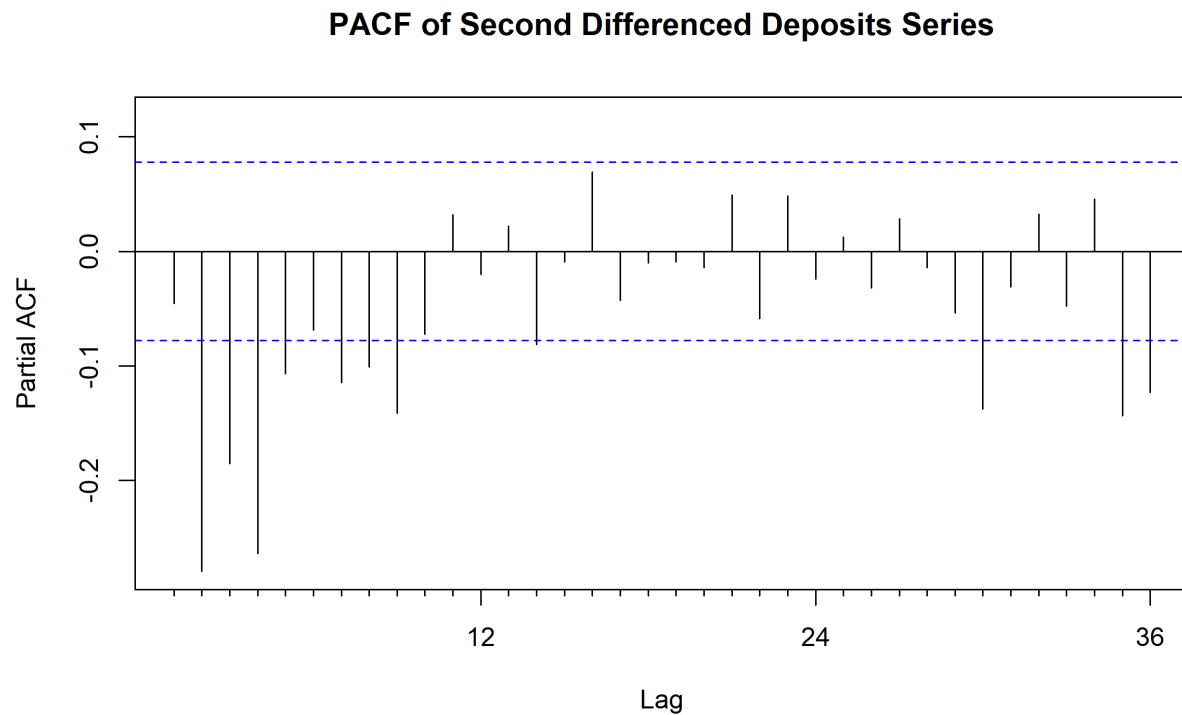


Figure 5. PACF of Second Differenced Deposits Series

Figure 5 shows the partial autocorrelation function of the second differenced series. The PACF displays significant spikes at lags 1 and 2, followed by a gradual decay. This pattern suggests a combination of autoregressive and moving average components rather than a purely autoregressive process. The seasonal spike at lag 12 supports inclusion of a seasonal AR component. Together with the ACF results, these patterns support the use of an ARIMA model with both AR and MA terms.

Compared to the first differenced series, the second differenced series exhibits a more stable partial autocorrelation structure with reduced persistence, further supporting its use for model estimation.

DATA TRANSFORMATION

Three transformations were performed to prepare the data for modeling:

- Missing value interpolation
- Evaluation of first and second differencing
- Training and validation data split

Missing Value Treatment

The dataset contained one missing value, which was replaced using linear interpolation prior to modeling. This ensured the time series remained complete and compatible with R time series modeling functions.

Differencing Decision

First differencing achieved stationarity based on the Augmented Dickey-Fuller test results. However, ACF and PACF analysis indicated remaining autocorrelation persistence after first differencing. To determine the appropriate level of differencing, ARIMA models with both first differencing ($d = 1$) and second differencing ($d = 2$) were estimated and compared. The model with second differencing produced lower validation errors and more appropriate residual diagnostics, including residuals consistent with white noise. Therefore, second differencing was retained based on superior predictive performance rather than stationarity alone.

Training and Validation Split

The data were divided chronologically into training and validation sets using an approximately 80/20 split. The training set was used to estimate all candidate models, while the validation set was used to evaluate predictive accuracy on unseen observations.

Variance Stabilization

Although increased variability is visible during the COVID-19 period, particularly in the residual diagnostics, overall model diagnostics indicated acceptable residual behavior without transformation, so a Box Cox transformation was not applied.

These transformations prepared the series for reliable model estimation and forecasting.

MOTIVATION BASED ON THEORETICAL CONSIDERATIONS FOR THE CHOICE OF TECHNIQUE(S)

Multiple model families were evaluated to ensure a robust comparison of forecasting performance. Mean, naïve, and seasonal naïve models provided baseline forecasts used to evaluate whether more complex models produced meaningful improvements in accuracy, since any useful forecasting model should outperform simple persistence forecasts.

Regression models were included to capture deterministic trend behavior and provide a transparent parametric baseline for comparison against more flexible time series approaches. ETS models were also evaluated because exponential smoothing methods are theoretically appropriate when recent observations carry greater predictive weight than older observations, which is reasonable for a long trending financial series.

ARIMA models were selected as the primary technique because they are designed to model autocorrelation commonly observed in financial time series. The deposit series exhibits both trend and autocorrelation, and seasonal patterns observed in the ACF and PACF plots supported the use of a seasonal ARIMA specification. Comparing multiple model families ensured an evidence-based model selection process consistent with standard forecasting methodology.

ITERATIVE RESULTS

Multiple Forecasting Techniques Evaluated

Several forecasting models were evaluated to identify the best performing approach. The models tested included Mean, Naïve, Seasonal naïve, Regression, ETS, and ARIMA models. These provided a range of benchmark and advanced forecasting approaches. Additional comparison between ARIMA models with first differencing ($d = 1$) and second differencing ($d = 2$) showed that the second differenced model provided significantly better validation performance and residual diagnostics.

Measuring Predictor Performance

Models were trained using the training dataset and evaluated on validation data to ensure fair comparison. Forecast accuracy was evaluated using **ME, RMSE, MAE, MPE, MAPE, MASE, ACF1, and Theil's U**.

Table 2. Forecast Accuracy Comparison

Model	Dataset	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Mean	Training Set	0.000	2730.664	2234.608	-82.057	111.206	9.403	0.992	NA
Mean	Validation Set	11412.931	11761.219	11412.931	75.205	75.205	48.022	0.984	82.081
Naïve	Training Set	19.918	32.986	22.069	0.564	0.655	0.093	0.373	NA
Naïve	Validation Set	4253.641	5115.147	4253.641	25.587	25.587	17.898	0.984	32.261
Seasonal Naïve	Training Set	237.577	310.936	237.658	6.541	6.544	1.000	0.985	NA
Seasonal Naïve	Validation Set	4512.630	5332.764	4512.630	27.391	27.391	18.988	0.984	33.792
Regression	Training Set	0.000	995.986	823.000	10.957	37.993	3.463	0.990	NA
Regression	Validation Set	5905.388	6317.816	5905.388	37.868	37.868	24.848	0.986	42.238
ETS	Training Set	1.469	21.354	12.764	0.039	0.378	0.054	0.041	NA
ETS	Validation Set	972.308	1560.093	1206.003	5.294	7.209	5.075	0.992	9.835
ARIMA d = 1	Training Set	0.090	22.444	13.480	-0.125	0.436	0.057	-0.014	NA
ARIMA d = 1	Validation Set	2895.100	3607.432	2895.100	17.138	17.138	12.182	0.987	22.586
ARIMA d = 2	Training Set	1.578	21.234	12.654	0.045	0.396	0.053	-0.014	NA
ARIMA d = 2	Validation Set	961.235	1550.264	1199.818	5.224	7.180	5.048	0.992	9.778

Table 2 presents the forecast accuracy metrics for all forecasting models evaluated on both the training and validation datasets. The benchmark models (Mean, Naïve, and Seasonal Naïve) produced higher validation errors, confirming that simple persistence forecasts are inadequate for this series. ETS and the ARIMA models produced the strongest overall performance, with the ARIMA $d = 2$ specification achieving the lowest validation MASE (5.048), only marginally outperforming ETS (5.075). Because ARIMA explicitly models autocorrelation and seasonal structure identified in the diagnostic analysis, the ARIMA $d = 2$ specification was selected as the final model.

The substantial increase in validation errors relative to training errors reflects the structural break associated with the COVID-19 period and the subsequent normalization in commercial bank deposits. Because the validation period contains unusual economic conditions not fully represented in the earlier training data, forecast uncertainty is naturally higher during this period. This behavior is common in financial time series affected by major macroeconomic shocks and should be considered when interpreting long-term forecast confidence.

How Overfitting Was Avoided

Overfitting was reduced through four steps. Data was split into training and validation sets to test performance on unseen observations. AICc based selection penalized excessive model complexity. Residual diagnostics confirmed no remaining structure, and final selection was based on validation accuracy rather than training fit.

In addition, model comparison between ARIMA specifications with first differencing ($d = 1$) and second differencing ($d = 2$) ensured that the selected model avoided overfitting by choosing the level of differencing that provided the best out-of-sample performance.

Model Tuning Actions Performed

The final model ARIMA(2,2,1)(1,0,1)[12] was selected after refinement. The AR(2) terms capture short term dynamics, while the MA(1) term captures short term error correction. Second differencing ($d = 2$) was selected after comparative evaluation, as it produced lower validation errors and improved residual diagnostics relative to first differencing. The seasonal AR(1) and seasonal MA(1) terms capture yearly persistence and seasonal error correction. No seasonal differencing was required ($D = 0$).

Model selection used the full `auto.arima()` search supported by ACF and PACF analysis.

Iterative Model Improvement

Initial models included benchmark approaches. ETS and ARIMA performed best among the tested models. ARIMA orders were refined using ACF and PACF analysis together with `auto.arima()` model selection. This iterative refinement resulted in the final ARIMA(2,2,1)(1,0,1)[12] specification. Additional comparison of models with different levels of

differentencing confirmed that the second differenced ARIMA model provided superior predictive performance.

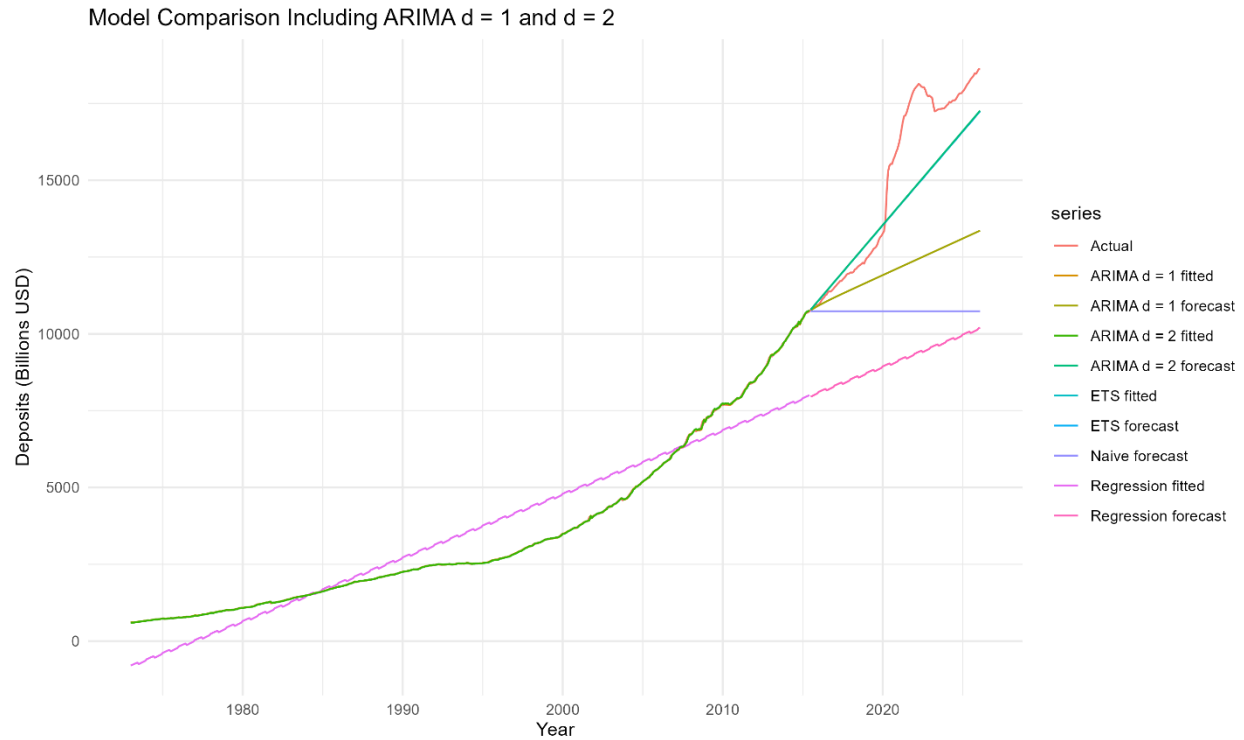


Figure 6. Model Comparison Including ARIMA d = 1 and d = 2

Figure 6 compares fitted values and validation forecasts across model families, including ARIMA models with first differencing and second differencing. The second-differenced ARIMA forecast tracks the validation period more closely than the first-differenced ARIMA forecast. This supports the decision to retain second differencing despite first differencing achieving stationarity in the ADF test.

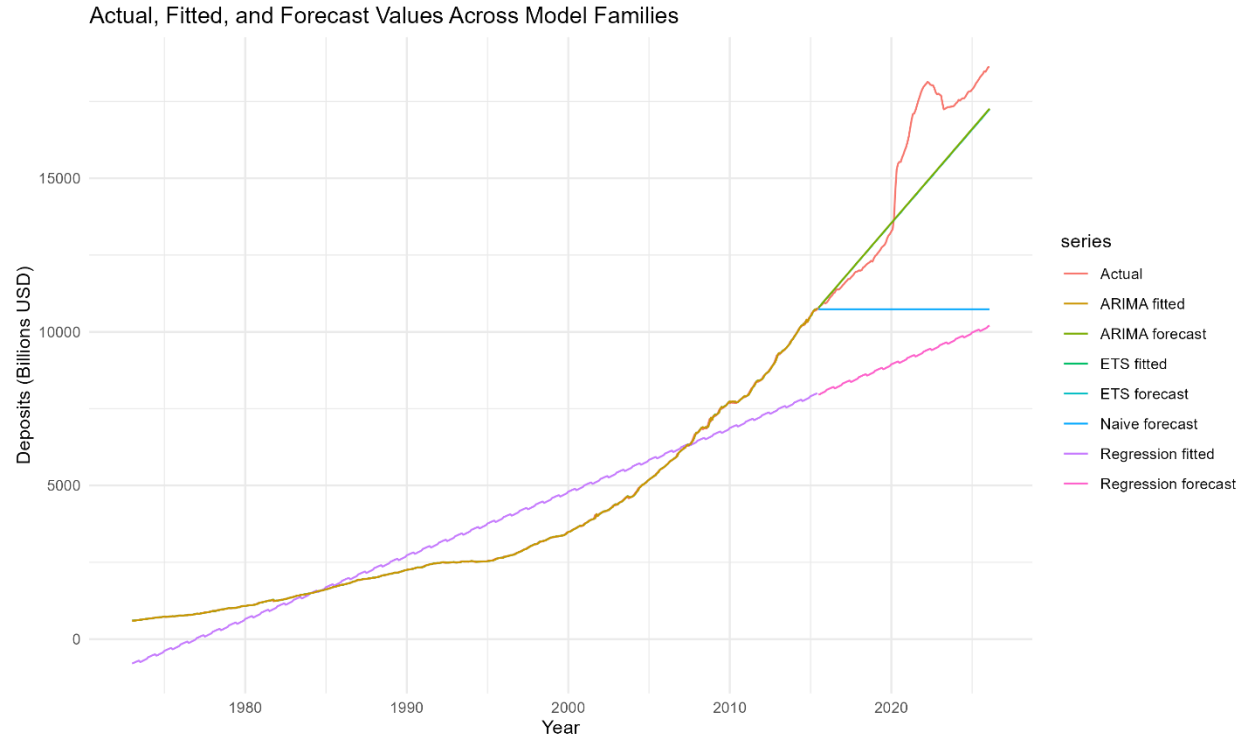


Figure 7. Forecast Model Comparison

Figure 7 compares the final selected ARIMA model with ETS, regression, and benchmark forecasting approaches. The ARIMA model follows the observed series closely and performs comparably to ETS while outperforming benchmark forecasting approaches, supporting its selection as the final forecasting model.

DESCRIPTION OF RESULTS

The final model selected was $ARIMA(2,2,1)(1,0,1)[12]$, based on comparative evaluation of models with different levels of differencing. Although first differencing achieved stationarity, the second differenced model produced lower validation errors and more appropriate residual diagnostics, supporting its selection as the final model.

Residual Diagnostics

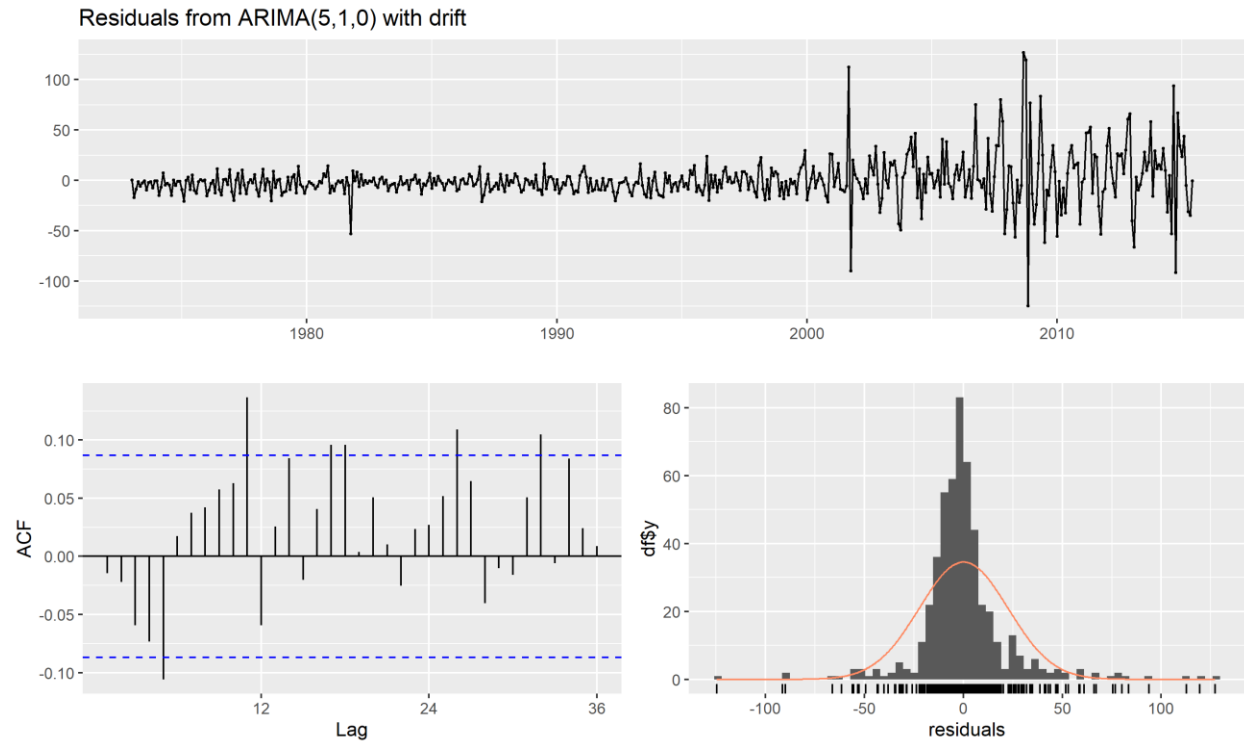


Figure 8. Residual Diagnostics for ARIMA(5,1,0)

Figure 8 shows residual diagnostics for the ARIMA(5,1,0) model. The residual time plot displays fluctuations around zero but with increasing variance over time, particularly in later periods. The residual ACF shows several autocorrelations outside the confidence bounds, indicating remaining dependence in the residuals. The histogram appears normal but with some dispersion. Overall, these diagnostics suggest that the ARIMA(5,1,0) model does not fully capture the underlying structure of the data and is therefore not suitable as the final model.

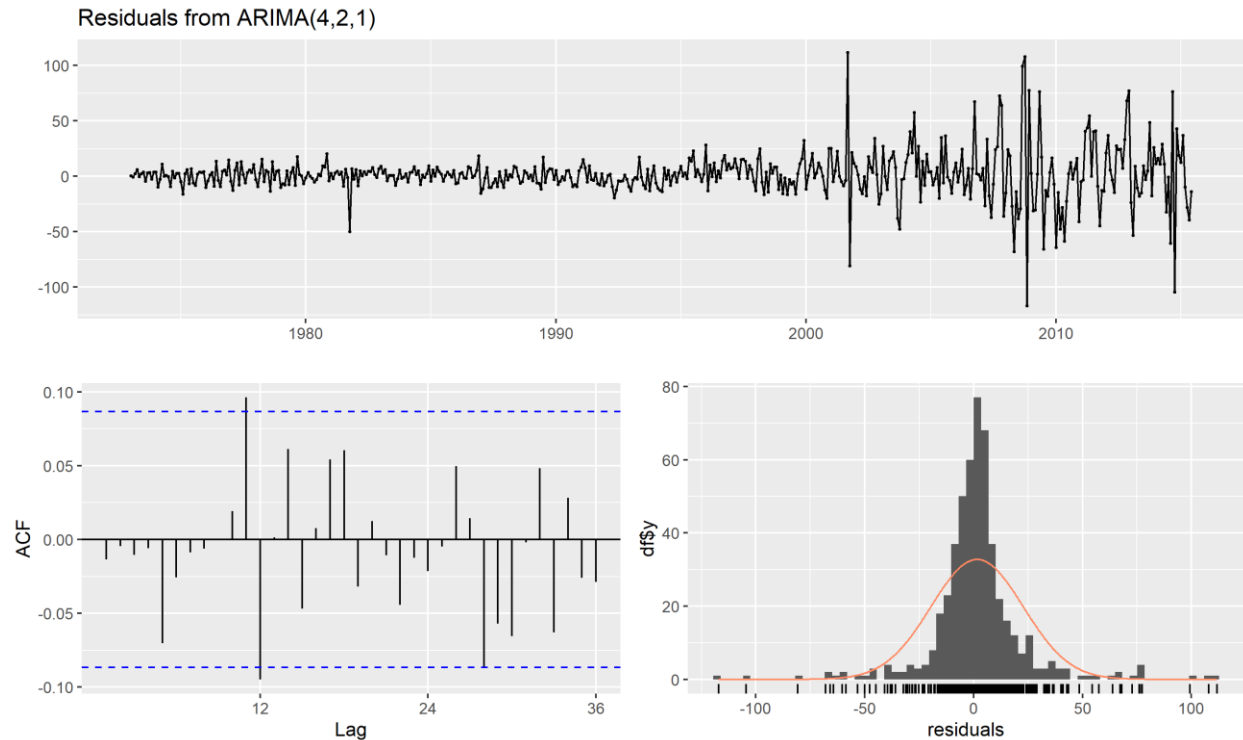


Figure 9. Residual Diagnostics for ARIMA(4,2,1)

Figure 9 presents residual diagnostics for the ARIMA(4,2,1) model, which was selected by `auto.arima()` on the training subset. The residual time plot shows random fluctuations around zero, but with noticeable variability in later periods. The residual ACF indicates that most autocorrelations fall within the confidence bounds, suggesting limited remaining dependence. The histogram is normally distributed.

While this model provides a reasonable fit, it was not selected as the final model because the model refit on the full dataset produced a different specification with improved performance. The final model incorporated seasonal AR and MA terms, suggesting that the complete dataset provided stronger evidence of annual seasonal structure than the training subset alone.

Residual Diagnostics Final Model

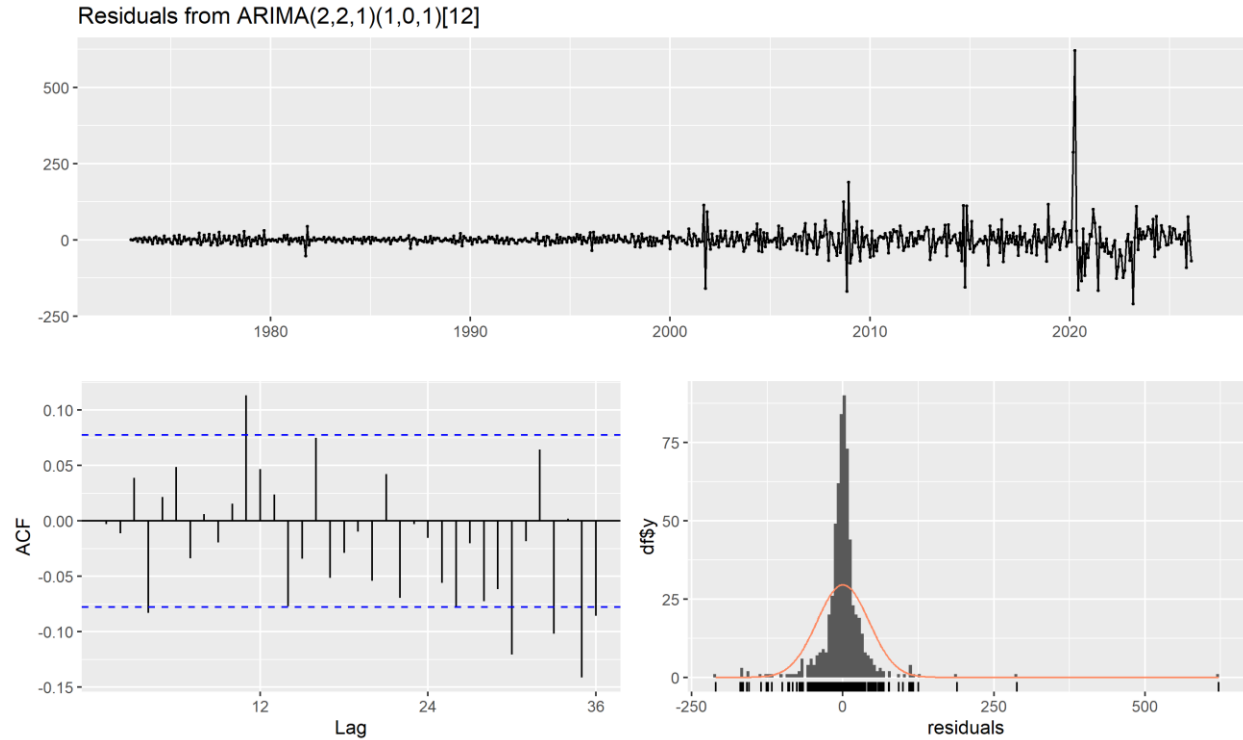


Figure 10. Final ARIMA Model Residual Diagnostics

Figure 10 shows residual diagnostics for the final ARIMA(2,2,1)(1,0,1)[12] model. The residual time plot shows mostly random fluctuations around zero with no clear systematic pattern. A noticeable spike occurs around 2020, corresponding to COVID-related changes in commercial bank deposits, which represents an external structural shock rather than model inadequacy. The residual ACF shows only minor remaining autocorrelation within acceptable bounds. The histogram indicates that residuals are normally distributed. These results confirm that the final model satisfies the key assumptions required for reliable forecasting.

Prediction of Future Events

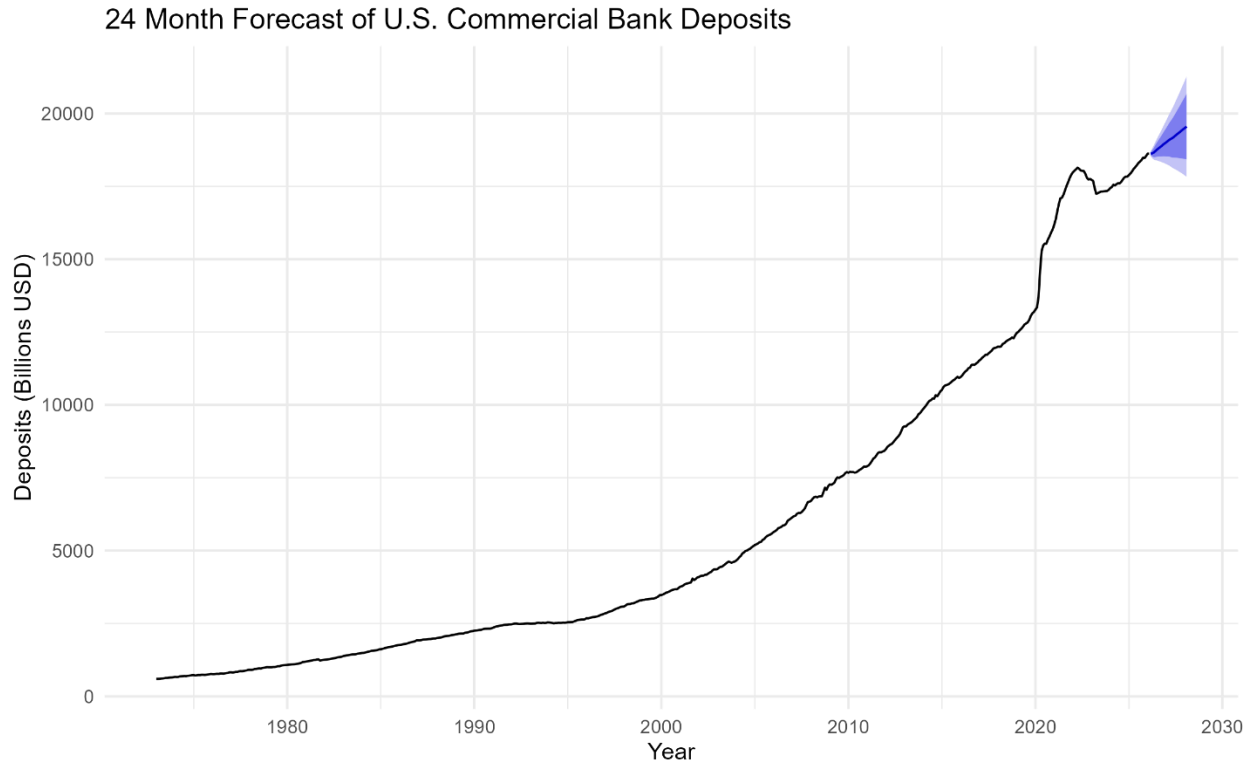


Figure 11. 24 Month Forecast of U.S. Commercial Bank Deposits

Figure 11 presents the 24-month forecast of U.S. commercial bank deposits based on the final ARIMA model. The forecast indicates continued growth in deposits, consistent with the long-term upward trend observed in the historical data. The shaded regions represent the 80% and 95% prediction intervals, which widen as the forecast horizon increases, reflecting growing uncertainty over time. This behavior is typical in time series forecasting and highlights the increasing difficulty of predicting further into the future.

Table 3. 24 Month Forecast of U.S. Commercial Bank Deposits

Values are in billions of U.S. dollars.

Month	Forecast	Lower 80%	Upper 80%	Lower 95%	Upper 95%
Mar 2026	18619	18564	18675	18535	18704
Apr 2026	18634	18521	18746	18462	18805
May 2026	18663	18501	18825	18415	18910
Jun 2026	18712	18507	18916	18398	19025
Jul 2026	18759	18515	19004	18385	19133

Aug 2026	18804	18520	19087	18370	19237
Sep 2026	18844	18521	19167	18350	19338
Oct 2026	18886	18522	19250	18330	19443
Nov 2026	18934	18528	19340	18314	19555
Dec 2026	18975	18527	19424	18289	19661
Jan 2027	19013	18521	19506	18260	19767
Feb 2027	19058	18521	19596	18237	19880
Mar 2027	19099	18517	19681	18209	19989
Apr 2027	19130	18503	19757	18172	20089
May 2027	19161	18488	19833	18132	20189
Jun 2027	19204	18485	19924	18105	20304
Jul 2027	19249	18482	20016	18076	20422
Aug 2027	19293	18478	20109	18046	20540
Sep 2027	19335	18470	20200	18012	20658
Oct 2027	19378	18462	20293	17978	20778
Nov 2027	19425	18458	20391	17946	20903
Dec 2027	19466	18447	20485	17908	21024
Jan 2028	19506	18434	20578	17866	21145
Feb 2028	19550	18424	20676	17828	21272

Table 3 presents the numerical forecast values corresponding to Figure 11. The point forecasts indicate a steady increase in U.S. commercial bank deposits from approximately 18.619 trillion in March 2026 to about 19.550 trillion in February 2028 over the 24-month horizon. The lower and upper bounds of the 80% and 95% prediction intervals provide a range of plausible future values and widen overtime, reflecting increasing uncertainty as the forecast horizon extends. These results suggest continued deposit growth while recognizing the uncertainty inherent in long-term financial forecasting.

CONCLUSION

This project developed and validated a univariate time series forecasting model for U.S. commercial bank deposits by systematically comparing six forecasting model families using held-out validation data.

ARIMA(2,2,1)(1,0,1)[12] provided the best predictive accuracy based on validation MASE, directly answering the first research question. The 24-month forecast from March 2026 to February 2028 shows continued moderate deposit growth, addressing the second research question. Model comparison results demonstrated clear performance differences across families, addressing the third research question.

From a financial decision-making perspective, the results suggest stable deposit growth conditions supporting continued lending capacity and liquidity planning for commercial banks. The widening prediction intervals highlight the importance of incorporating uncertainty into liquidity risk management decisions. Although the long-term historical trend supports continued deposit growth, recent periods of deposit stabilization following post-pandemic monetary tightening suggest that future growth may occur more slowly than projected under historical trend continuation assumptions.

Limitations include reliance on historical deposit data only. Future research could incorporate macroeconomic variables such as interest rates, inflation, or policy indicators to improve forecasting performance.

ESTIMATED AND ACTUAL TIME SPENT IN HOURS ON EACH TASK LISTED IN THE WORK PLAN (APPENDIX)

Week	Date	Category	Task	Estimated Hours	Actual Hours
9	9 Mar	Cleaning, preparing, and getting familiar with my data	Explore dataset	0.75	0.75
			Clean data by handling missing values, anomalies, and outliers	1.50	1.25
			Run and analyze descriptive statistics	3.00	2.75
			Plot time series and visually inspect trend and seasonality	1.50	1.50
10	23 Mar	Majority of the statistical and programming work	Test for stationarity and apply differencing if necessary	1.50	1.75
			Plot ACF and PACF to identify candidate ARIMA model orders	1.50	1.50
			Split data into training and validation sets	1.50	1.25
			Fit ARIMA and alternative forecasting models	4.50	4.75
			Evaluate model performance using forecast accuracy metrics	3.00	3.25
			Conduct residual diagnostics to confirm model adequacy	1.50	1.50

			Document model performance and results	1.50	1.75
11	30 Mar	Finalizing model and starting my data analysis paper	Finish final data analysis to select the best model	1.50	1.50
			Write first draft of the project data analysis	3.00	3.25
12	6 Apr	Finalizing my data analysis paper	Finalize data analysis project	3.00	3.25
13	13 Apr	Preparing to record my presentation	Create PowerPoint with relevant visualizations	3.00	2.75
			Write a script of my presentation	3.00	2.75
14	20 Apr	PowerPoint presentation	Record presentation and PowerPoint	3.00	4.00
16	7 May	Revise and submit final project	Make revisions on project written report based on feedback	2.00	4.00

Total Estimated Hours: 40.25

Total Actual Hours: 43.50